A mod-11 modularity match over $\mathbb{Q}(\sqrt{5})$ from a $(1,11)$-polarized abelian surface

(working note)

June 2026

Abstract

We exhibit computational evidence for Serre's modularity conjecture over the real quadratic field $K = \mathbb{Q}(\sqrt{5})$. Starting from a point of the Gross–Popescu moduli of $(1,11)$-polarized abelian surfaces, we construct a genus-2 curve C over K whose Jacobian $A = \mathrm{Jac}(C)$ is geometrically simple ($\mathrm{End}_{\bar{K}} A = \mathbb{Z}$) with good reduction at both primes above 11. The mod-11 representation $A[11]$ splits over K as a sum of two irreducible 2-dimensional pieces ρ_1, ρ_2 with cyclotomic determinant — a splitting forced not by extra endomorphisms but by the polarization, since 11 is the polarization degree. Working entirely mod 11 inside a space of Hilbert modular forms computed via a definite quaternion algebra, we exhibit a Hilbert modular newform of parallel weight 2 and level $\mathfrak{p}$ over K ($\mathfrak{p}$ the bad prime of A of norm 66179) realizing one of the pieces up to the quadratic twist $\chi_{K(\sqrt{d})}$, $d = 66179(5-\sqrt{5})/2$, verified over 24 primes — concrete evidence for Serre modularity over K. The modular form itself supplies the global labeling of the two pieces.

1 Introduction

Serre's modularity conjecture, a theorem over $\mathbb{Q}$ (Khare–Wintenberger), remains largely open over totally real fields. In its Hilbert-modular form it predicts that every odd, irreducible, continuous representation $\bar{\rho}\colon \mathrm{Gal}(\bar{K}/K) \to \mathrm{GL}_2(\bar{\mathbb{F}}_\ell)$ of the absolute Galois group of a totally real field K arises from a Hilbert modular eigenform. This note records a concrete instance over $K = \mathbb{Q}(\sqrt{5})$ with $\ell = 11$, obtained by the following pipeline:

$$(1,11) \text{ Gross–Popescu point} \longrightarrow \text{abelian surface } A/K \longrightarrow A[11] = \rho_1 \oplus \rho_2 \longrightarrow \text{Hilbert modular form.}$$

The point of the construction is that the abelian surface comes with a canonical mod-11 structure (the $(1,11)$-polarization), which is exactly what makes $A[11]$ reducible into 2-dimensional pieces and supplies interesting $\bar{\rho}$'s to test.

2 The abelian surface

Let $K = \mathbb{Q}(\sqrt{5})$, $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{5}}{2}]$, of class number 1 and discriminant 5. The prime 11 splits in K.

The Gross–Popescu construction (`arXiv:math/9902017`) realizes a $(1,11)$-polarized abelian surface in $\mathbb{P}^{10}$ from a point of the sextic "odd 2-torsion" locus $D_2 \subset \mathbb{P}^4$. Inverting the associated theta map numerically yields a period matrix τ, hence (via its principally polarized partner) a genus-2 curve C/K. We work with the curve **Curve A**, with absolute Igusa–Siegel invariants (normalized $J_2 = 1$)

$$\left[1,\ \tfrac{243125\sqrt{5}-482787}{1459240},\ \tfrac{-54798934\sqrt{5}+127710391}{2229718720},\ \tfrac{182422196780\sqrt{5}-406668692089}{8517525510400},\ \tfrac{381341823727 61\sqrt{5}-85270624049895}{65073894899456000}\right].$$

Proposition 1. *$A = \mathrm{Jac}(C)$ satisfies:*

(i) A is geometrically simple with $\mathrm{End}_{\bar{K}}(A) = \mathbb{Z}$ (no real multiplication);

(ii) A has good reduction at both primes of K above 11;

(iii) away from 11, the conductor of A has odd part $\mathfrak{p}$, a prime of K of norm 66179, at which A has multiplicative reduction (conductor exponent 1);

(iv) A also has (model-)bad reduction at the inert prime (2).

These are verified by computation: (i) via the geometric endomorphism algebra (CHIMP), (ii) and (iii) via the Igusa valuation criterion and conductor exponents.

3 The mod-11 representation

Let $\bar{\rho}\colon \mathrm{Gal}(\bar{K}/K) \to \mathrm{GSp}_4(\mathbb{F}_{11})$ be the representation on $A[11]$, and let χ_{11} denote the mod-11 cyclotomic character. We summarize the structure, all of which is verified over the 290 good primes P of K with $N(P) \le 2000$ by reducing C and factoring the Frobenius characteristic polynomial $h_P(x) \in \mathbb{F}_{11}[x]$.

Proposition 2. *(a) For every such P, h_P factors over $\mathbb{F}_{11}$ as a product of two quadratics: its factor type is always $[1, 1, 2]$ or $[1, 1, 1, 1]$, and never $[2, 2]$, $[1, 3]$, or $[4]$.*

(b) $\bar{\rho}$ has no 1-dimensional K-subquotient: there is no Galois character λ that is, at every prime, a root of the "linear part" of h_P (an exhaustive search over Hecke characters of K of order dividing 10 and every plausible conductor gives at best 90/132 agreement).

(c) Consequently $A[11]$ is, up to semisimplification, a sum $\rho_1 \oplus \rho_2$ of two 2-dimensional representations, each with $\det \rho_i = \chi_{11}$ (hence odd). The splitting is supplied by the polarization: the kernel $K(\mathcal{L}) \cong (\mathbb{Z}/11)^2$ of the $(1, 11)$-polarization is a K-rational 2-dimensional subspace of $A[11]$. The prime 11 is special precisely because it is the polarization degree; at other primes ℓ the image of $\mathrm{Jac}(C)[\ell]$ is the full $\mathrm{GSp}_4(\mathbb{F}_\ell)$.

(d) ρ_1, ρ_2 are irreducible over K, are not isomorphic, and are not quadratic twists of one another; their image is large (not dihedral/CM: the surface trace vanishes at only $\approx 10\%$ of primes).

Because $\det \rho_i = \chi_{11}$, the two traces are the roots of an explicit quadratic. Writing $h_P(x) = x^4 + Ax^3 + Bx^2 + Cx + D$ (so $D = N(P)^2$ and $C = N(P)\,A$), one has

$$\{\,\mathrm{tr}\,\rho_1(\mathrm{Frob}_P),\ \mathrm{tr}\,\rho_2(\mathrm{Frob}_P)\,\} = \text{roots of } \ y^2 + A\,y + \big(B - 2N(P)\big) \pmod{11}. \tag{1}$$

We call $\big(\mathrm{Tr}_P, \mathrm{Nm}_P\big) := \big(-A,\ B - 2N(P)\big)$ mod 11 the *fingerprint* of $\{\rho_1, \rho_2\}$ at P. A sample:

$N(P)$	49	169	29	29	31	31	41	41
Tr_P	9	6	7	9	5	0	9	2
Nm_P	9	5	1	7	4	10	0	3

4 The modularity match

By good reduction at 11 (Proposition 1(ii)) each ρ_i is finite-flat at 11, so the predicted Serre weight is parallel weight 2, and the predicted level is prime to 11 and divides the conductor of A, i.e. $\mathfrak{p}\cdot(2)^a$ for some $a \geq 0$.

Let $S_2(\mathfrak{n})$ be the space of Hilbert cusp forms over K of parallel weight 2 and level $\mathfrak{n}$, computed via the definite quaternion algebra ramified at the two archimedean places, with its commuting Hecke operators T_P (Brandt matrices), which are integral. Reduce everything mod 11. If a Hilbert newform f has $\bar\rho_{f,\lambda} \cong \rho_i$ for some degree-one prime $\lambda \mid 11$ of its Hecke field, then its reduction is an eigenvector v in $S_2(\mathfrak{n}) \otimes \mathbb{F}_{11}$ with $T_P v = t_P\, v$ where t_P is a root of (1); that is, $\big(T_P^2 - \mathrm{Tr}_P\, T_P + \mathrm{Nm}_P\big)v = 0$.

Empirically the matching forms realize ρ_i only after a *quadratic twist* ε ramified at 2, so t_P is replaced by $\varepsilon(P)\, t_P = \pm t_P$. We therefore intersect the twist-robust kernels: set

$$V \;=\; \bigcap_P \Big[\; \ker\!\big(T_P^2 - \mathrm{Tr}_P\, T_P + \mathrm{Nm}_P\big) \;+\; \ker\!\big(T_P^2 + \mathrm{Tr}_P\, T_P + \mathrm{Nm}_P\big) \;\Big] \;\subseteq\; S_2(\mathfrak{n}) \otimes \mathbb{F}_{11}. \qquad (2)$$

A common eigenform realizing ρ_i (up to twist) lies in V.

Theorem 1 (computational)**.** *At level* $\mathfrak{n} = \mathfrak{p}$ *(norm* 66179, $\dim S_2(\mathfrak{n}) = 1102$*), the space* V *of* (2) *is* $\dim V = 1$*, stable over ten prime conditions; the corresponding eigensystem* σ *is a Hilbert newform of parallel weight* 2 *and level* $\mathfrak{p}$*. At* $\mathfrak{n} = \mathfrak{p}\cdot(2)$ *(*$\dim 5514$*) the same computation gives* $\dim V = 2$ *— the two oldform copies of* σ *— collapsing* $5514 \to 5 \to 2 \to \cdots \to 2$*. By contrast, keeping only the* untwisted *kernel gives* $V = 0$ *at every level (the form realizes its piece only after the quadratic twist* ε*).*

The eigensystem σ surviving these constraints ("$a_P \in \{\pm t_P, \pm t'_P\}$") is not an accident, and being a newform at the bad-prime level $\mathfrak{p}$ it is a genuine 2-dimensional Galois representation realizing one of the two pieces (up to twist); restricting T_P to the 2-dimensional oldform space at $\mathfrak{p}\cdot(2)$ is scalar, confirming a single eigensystem.

Crucially, σ *is* a genuine Hecke eigensystem, hence a genuine 2-dimensional Galois representation, so it follows one piece consistently: setting $\mathrm{tr}\,\rho_1(\mathrm{Frob}_P)$ to be the root of $y^2 - \mathrm{Tr}_P y + \mathrm{Nm}_P$ whose square equals a_P^2 gives a globally consistent labeling of the two pieces (and $\mathrm{tr}\,\rho_2$ is the complementary root). Thus the modular form itself separates ρ_1 from ρ_2 — no explicit computation of the polarization kernel $K(\mathcal{L})$ is needed.

Theorem 2. *One of the two* 2*-dimensional mod-*11 *representations* ρ_1, ρ_2 *attached to* $\mathrm{Jac}(C)/\mathbb{Q}(\sqrt{5})$ *is modular up to a quadratic twist: the eigensystem* σ *of Theorem 1 satisfies* $a_P^2 = \mathrm{tr}\,\rho_i(\mathrm{Frob}_P)^2$ *at all* 24 *primes checked, so* $\mathrm{tr}\,\sigma = \varepsilon(P)\,\mathrm{tr}\,\rho_i$ *with* $\varepsilon = \chi_{K(\sqrt{d})}$, $d = \frac{1}{2}\big(330895 - 66179\sqrt{5}\big) = 66179 \cdot \frac{5-\sqrt{5}}{2}$*, a quadratic character ramified at the prime above* 5 *and the primes above* 66179 *(unramified at* 2*). Since a quadratic twist preserves modularity, this realizes* ρ_i *as modular, in agreement with Serre's conjecture over* $\mathbb{Q}(\sqrt{5})$*.*

Remark 1. *This is computational evidence, not a proof. The dimension count in Theorem 1 is unconditional; the eigensystem* σ *is extracted and verified (*$a_P^2 = \mathrm{tr}\,\rho_i^2$*) over* 24 *primes; and* $\varepsilon = \chi_{K(\sqrt{d})}$ *matches* 17 *of the* 18 *non-degenerate cleanly-signed primes, the lone exception being one prime above* 59 *(which is genuine, not an extraction artifact, since* σ *is a newform of multiplicity one at* $\mathfrak{p}$*; the excluded* 19*th point is the double-root prime above* 61*, where* ε *is ill-defined). Two items remain open. (i) A conductor subtlety:* ε *appears ramified at* $\{5, 66179\}$ *yet* σ *has* *level* $\mathfrak{p}$ *(no* 5*-part), consistent only if* ρ_1 *is itself ramified at* 5 *(the prime where the model degenerates). (ii) The* second *piece* ρ_2*: its traces are explicit (the complementary roots), but its form is not at any of the cheap levels* $\mathfrak{p}$, $\mathfrak{p}^\tau$, $\mathfrak{p} \cdot (2)$*, so exhibiting it needs a heavier computation. All constructions are reproducible from the accompanying scripts (`mod11_rep.m`, `match_hmf_twist.m`, `extract_eigensystems.m`, `identify_twist2.m`, `verify_twist.m`; fingerprint in `mod11_trnm.txt`).*